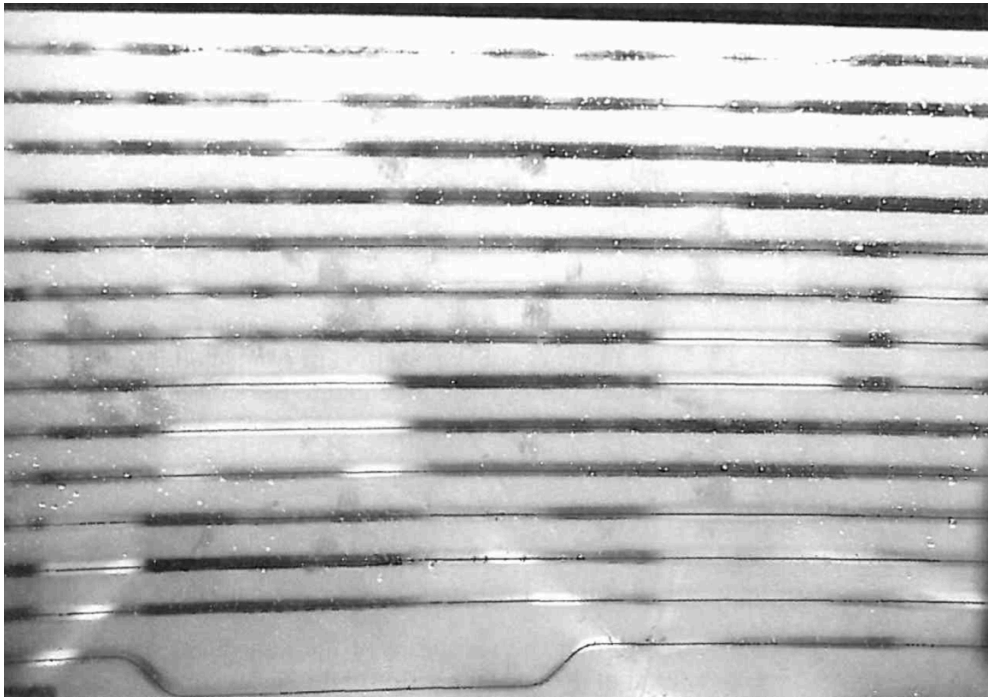
Simple Resistive Circuits

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Zhejiang University of Technology
September 2015

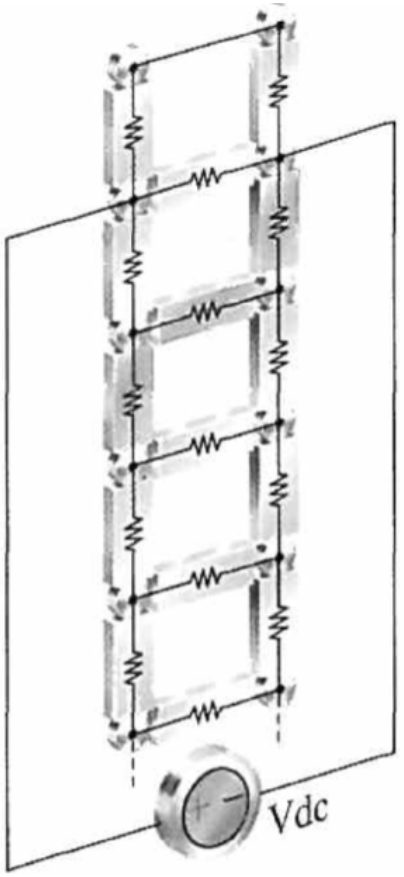
Structure

- Resistors in Series
- Resistors in Parallel
- The Voltage/Current-Divider Circuit
- Voltage/Current Division
- Measuring Voltage and Current
- Measuring Resistance-The Wheatstone Bridge
- Delta-to-Wye (Pi-to-Tee) Equivalent Circuit

A Rear Window Defroster

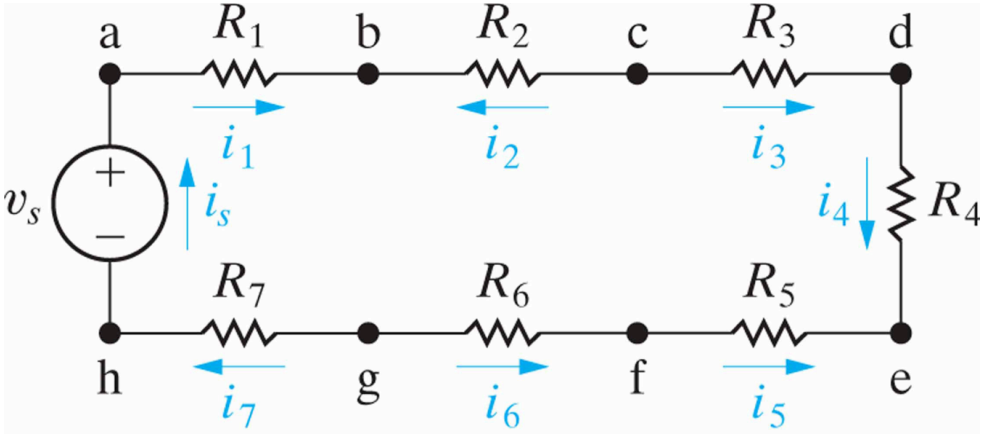


A Rear Window Defroster

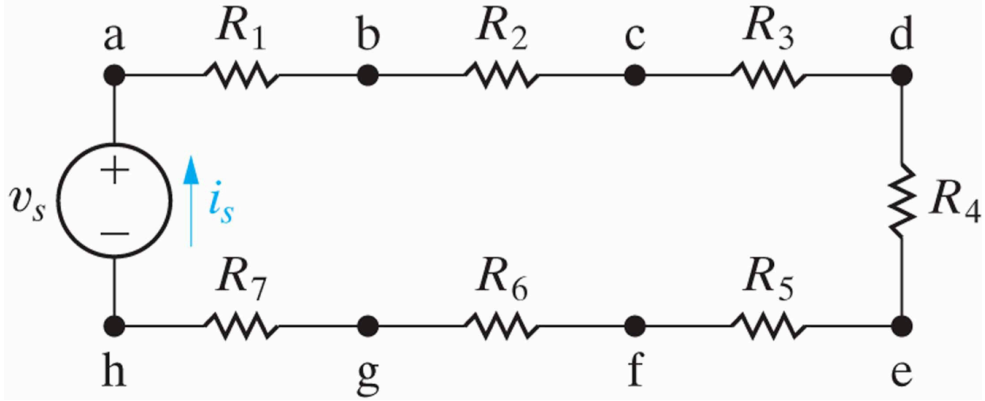


A resistive circuit

Resistors in Series



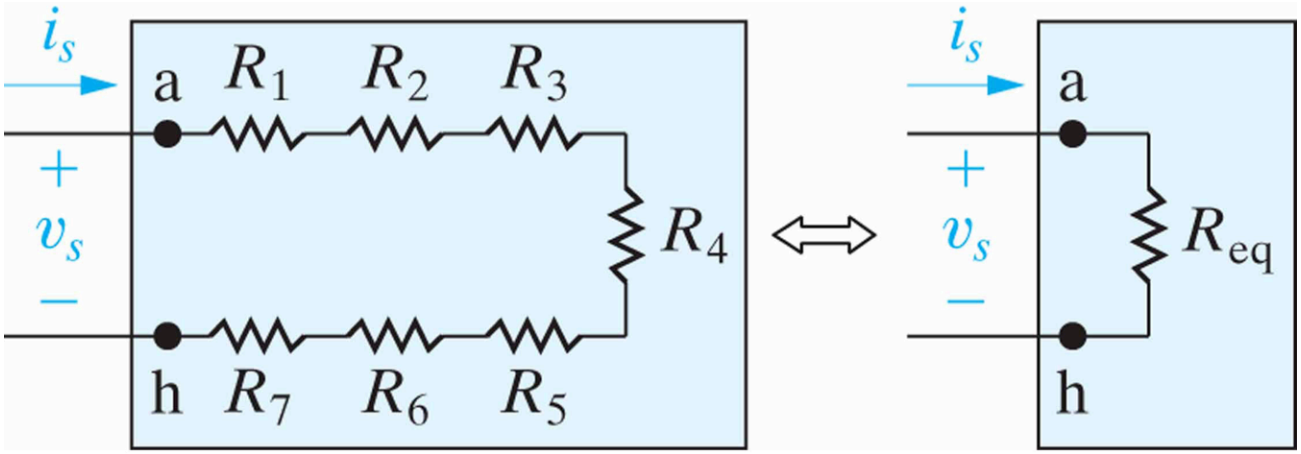
$$i_s = i_1 = -i_2 = i_3 = i_4 = -i_5 = -i_6 = i_7$$



$$v_s = i_s(R_1 + R_2 + R_3 + R_4 + R_5 + R_6 + R_7)$$

$$R_{eq} = R_1 + R_2 + R_3 + R_4 + R_5 + R_6 + R_7$$

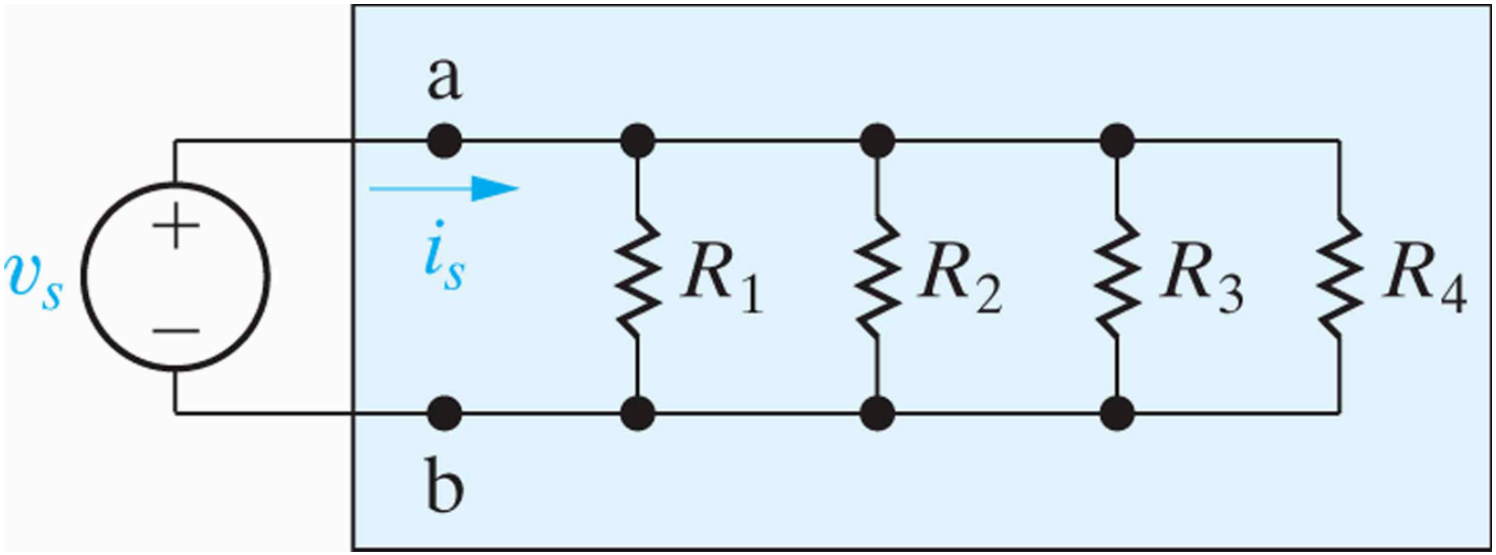
$$v_s = i_s R_{eq}$$



In general, if k resistors are connected in series, the equivalent single resistor has a resistance equal to the sum of the k resistances, or

$$R_{eq} = \sum_{i=1}^{k} R_i = R_1 + R_2 + \cdots + R_k$$

Resistors in Parallel



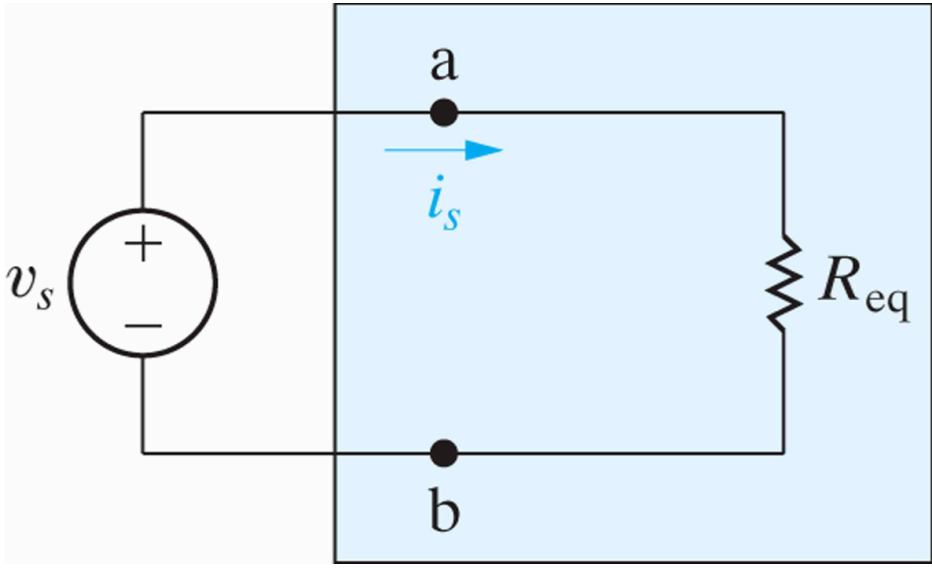
$$i_s = i_1 + i_2 + i_3 + i_4$$

$$i_1R_1 = i_2R_2 = i_3R_3 = i_4R_4 = v_s$$

$$i_1 = \frac{v_s}{R_1} \qquad i_2 = \frac{v_s}{R_2} \qquad i_3 = \frac{v_s}{R_3} \qquad i_4 = \frac{v_s}{R_4}$$

$$i_s = v_s\left(\frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3} + \frac{1}{R_4}\right)$$

$$\frac{i_s}{v_s} = \frac{1}{R_{eq}} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3} + \frac{1}{R_4}$$



$$\frac{1}{R_{eq}} = \sum_{i=1}^{k} \frac{1}{R_i} = \frac{1}{R_1} + \frac{1}{R_2} + \cdots + \frac{1}{R_k}$$

$$G_{eq} = \sum_{i=1}^{k} G_i = G_1 + G_2 + \cdots + G_k$$

Conductance

Typical Cases

- If k = 2, we have

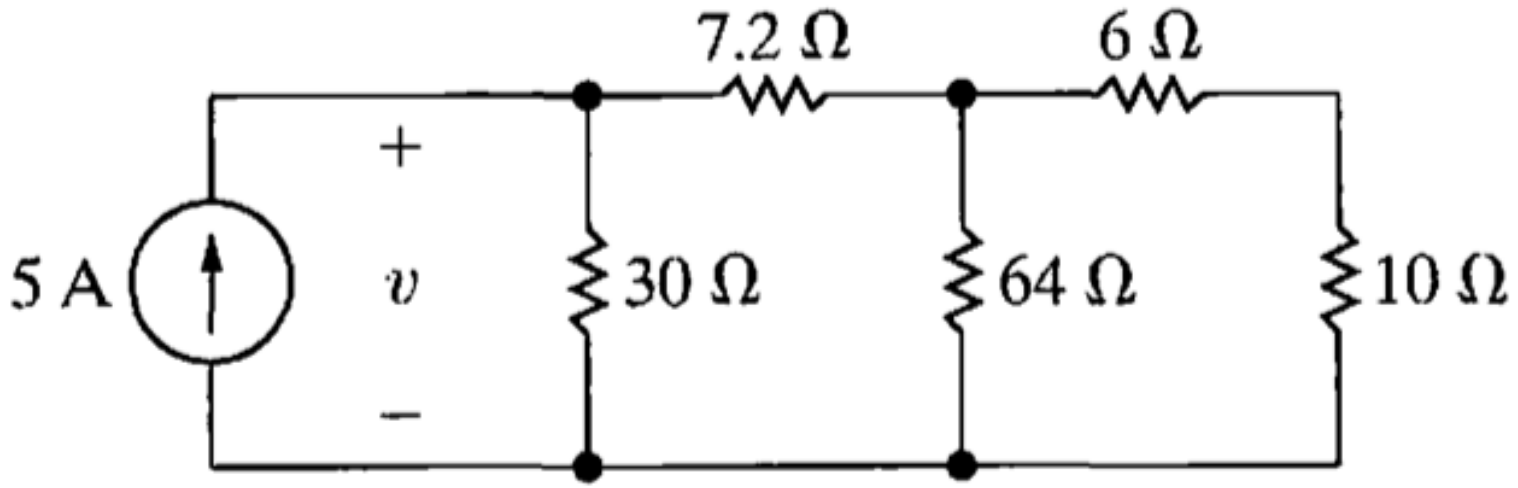
$$\frac{1}{R_{eq}} = \frac{1}{R_1} + \frac{1}{R_2} = \frac{R_2 + R_1}{R_1 R_2} \quad \Rightarrow \quad R_{eq} = \frac{R_1 R_2}{R_1 + R_2}$$

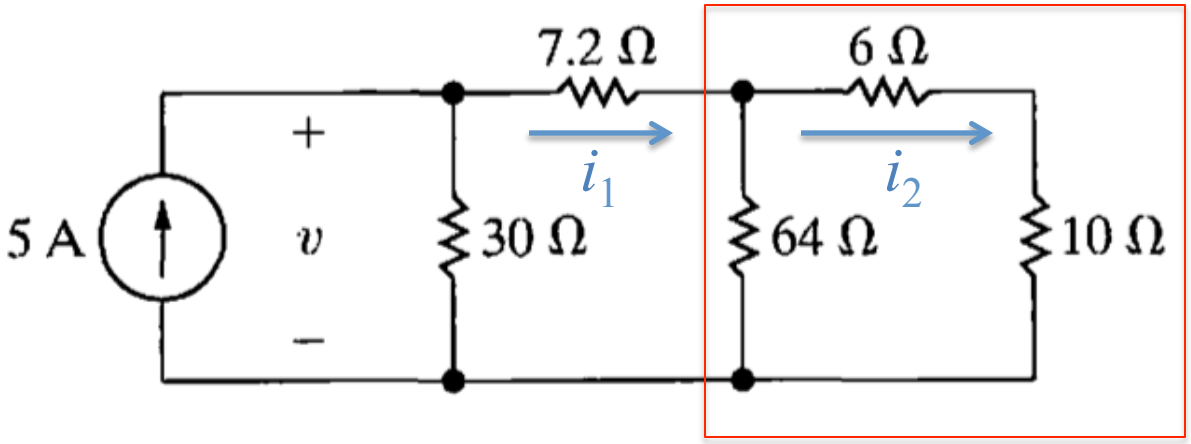
- If $R_1 = R_2 = \ldots\ldots = R_k = R$, we have

$$1/R_{eq} = k/R \quad \Rightarrow \quad R_{eq} = R/k$$

Example #1

- For the circuit shown, find (a) the voltage v, (b) the power delivered to the circuit by the current source, and (c) the power dissipated in the 10 Ω resistor.





R_1 v_1

$R_1 = (10+6)\times 64 / (10+6+64) = 12.8\ \Omega$

$R = (7.2+12.8)\times 30 / (7.2+12.8+30) = 12\ \Omega$

$v = 5R = 60\ \text{V}$

$i_1 = v / (7.2+R_1) = 60 / (7.2+12.8) = 3\ \text{A}$

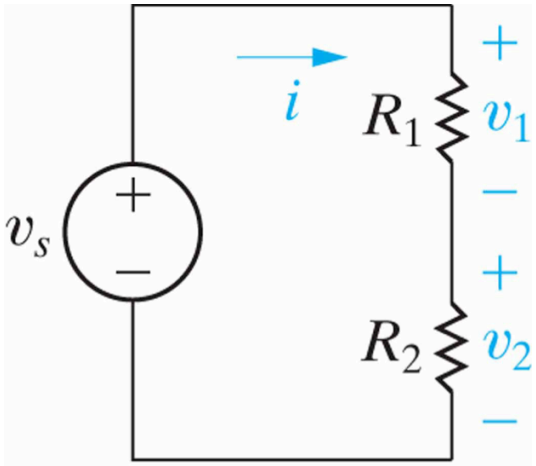
$v_1 = i_1 \times R_1 = 3\times 12.8 = 38.4\ \text{V}$

$i_2 = v_1 / (6+10) = 38.4 / 16 = 2.4\ \text{A}$

$p = i_2^2 \times 10 = 2.4\times 2.4\times 10 = 57.6\ \text{W}$

Solution for Example #1

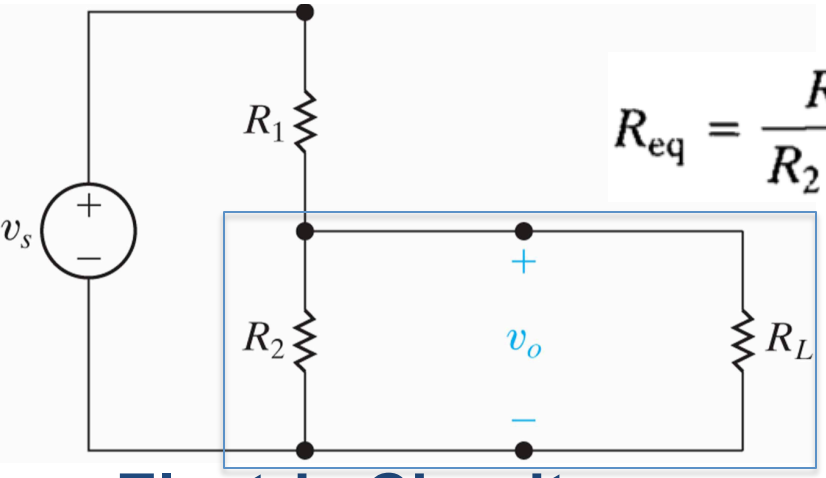
The Voltage-Divider Circuits



$$i = \frac{v_s}{R_1 + R_2}$$

$$v_1 = iR_1 = v_s \frac{R_1}{R_1 + R_2}$$

$$v_2 = iR_2 = v_s \frac{R_2}{R_1 + R_2}$$

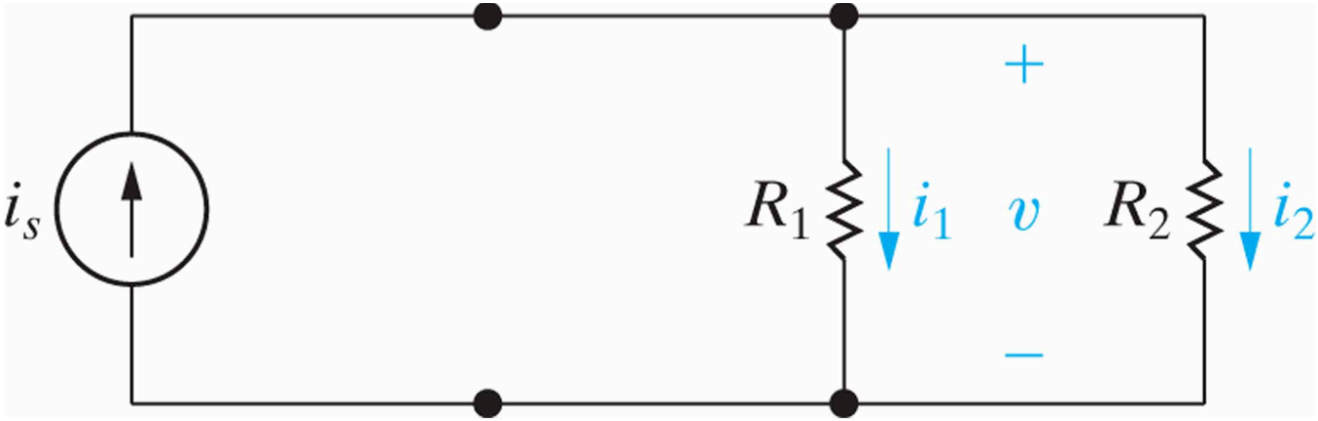


$$R_{eq} = \frac{R_2 R_L}{R_2 + R_L}$$

$$v_o = \frac{R_{eq}}{R_1 + R_{eq}} v_s$$

$$v_o = \frac{R_2}{R_1[1 + (R_2/R_L)] + R_2} v_s$$

The current-Divider Circuit



$$v = i_1 R_1 = i_2 R_2 = \frac{R_1 R_2}{R_1 + R_2} i_s$$

$$i_1 = \frac{R_2}{R_1 + R_2} i_s$$

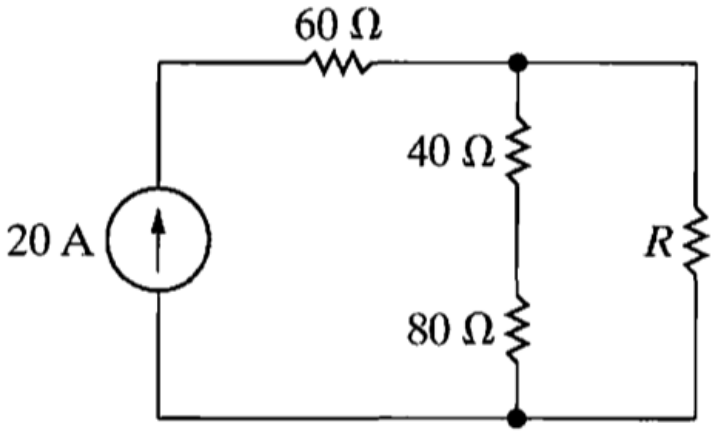
$$i_2 = \frac{R_1}{R_1 + R_2} i_s$$

Example #2

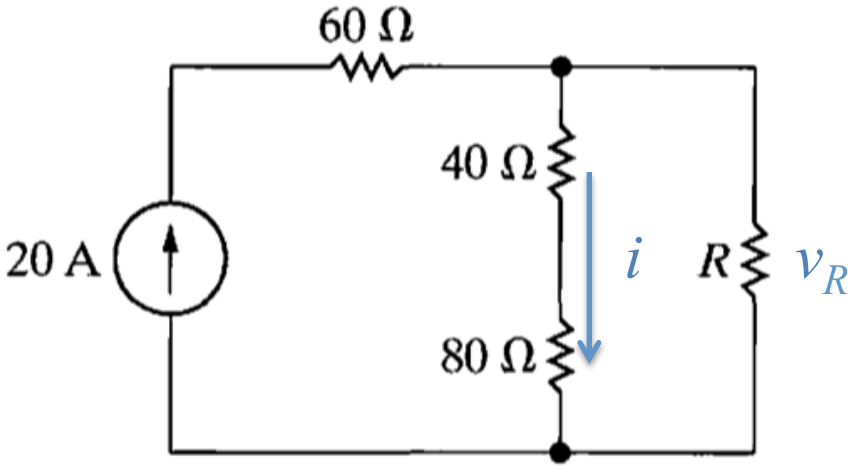
a) Find the value of R that will cause 4 A of current to flow through the 80 Ω resistor in the circuit shown.

b) How much power will the resistor R from part (a) need to dissipate?

c) How much power will the current source generate for the value of R from part (a)?



Solution for Example #2



For (a), we have

$i = 20R / (40+80+R) = 4$ A

$R = 30\ \Omega$

For (b), we have

$v_R = (40+80)i = 120\times4 = 480$ V

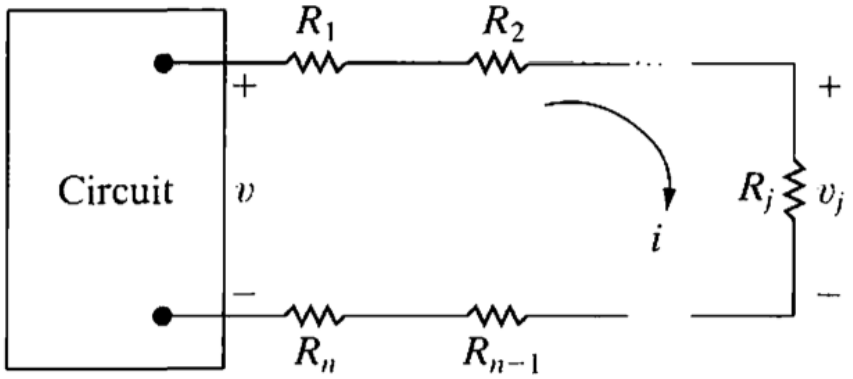
$p_R = v_R^2 / R = 480^2 / 30 = 7680$ W

For (c), we have

$v = 60\times20+v_R = 1200+480 = 1680$ V

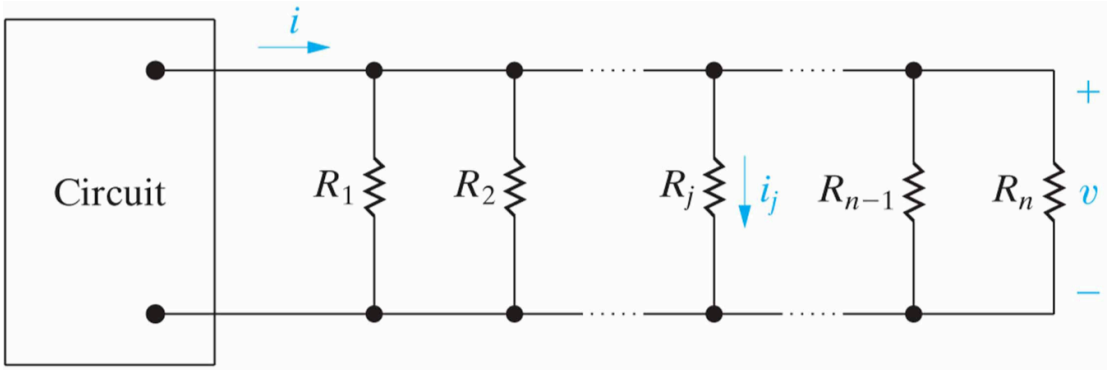
$p = 20v = 20\times1680 = 33600$ W

Voltage/Current Division



$$i = \frac{v}{R_1 + R_2 + \cdots + R_n} = \frac{v}{R_{eq}}$$

$$v_j = iR_j = \frac{R_j}{R_{eq}} v$$



$$v = i(R_1 \| R_2 \| \ldots \| R_n) = iR_{eq}$$

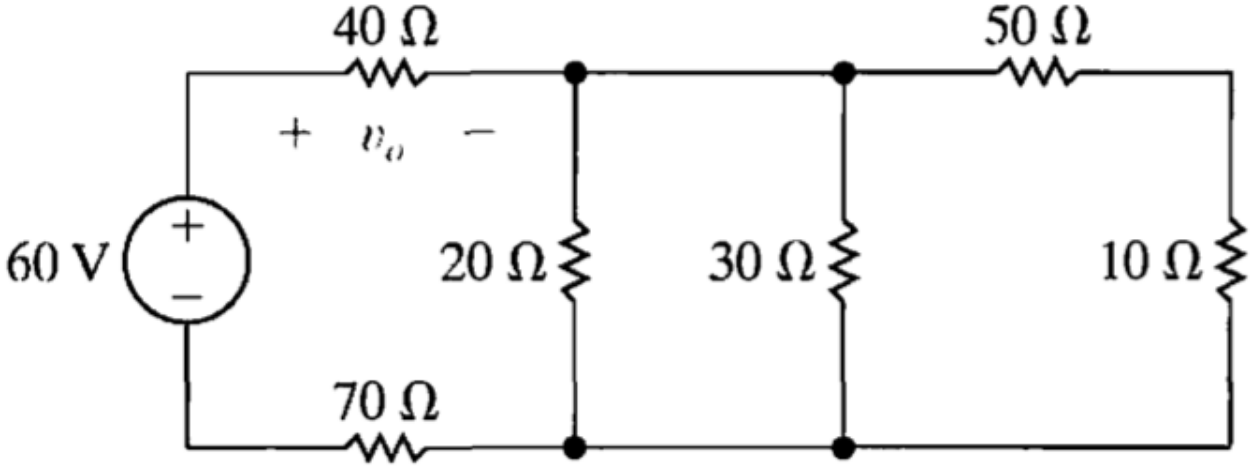
$$i_j = \frac{v}{R_j} = \frac{R_{eq}}{R_j} i$$

Example #3

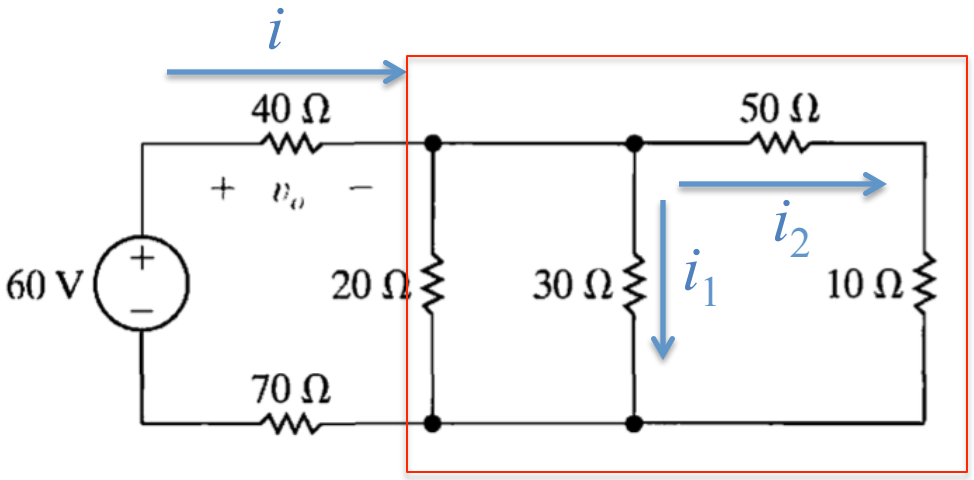
a) Use voltage division to determine the voltage v_0 across the 40 Ω resistor in the circuit shown.

b) Use v_0 from part (a) to determine the cur- rent through the 40 Ω resistor, and use this current and current division to calculate the current in the 30 Ω resistor.

c) How much power is absorbed by the 50 Ω resistor?



Solution for Example #3



For (a), we have

$R_{eq} = 40+(20\|30\|(50+10))+70$

$= 40+10+70=120\ \Omega$

$v_o = 60\times40 / R_{eq} = 20$ V

For (b), we have

$i = v_o / 40 = 20 / 40 = 0.5$ A

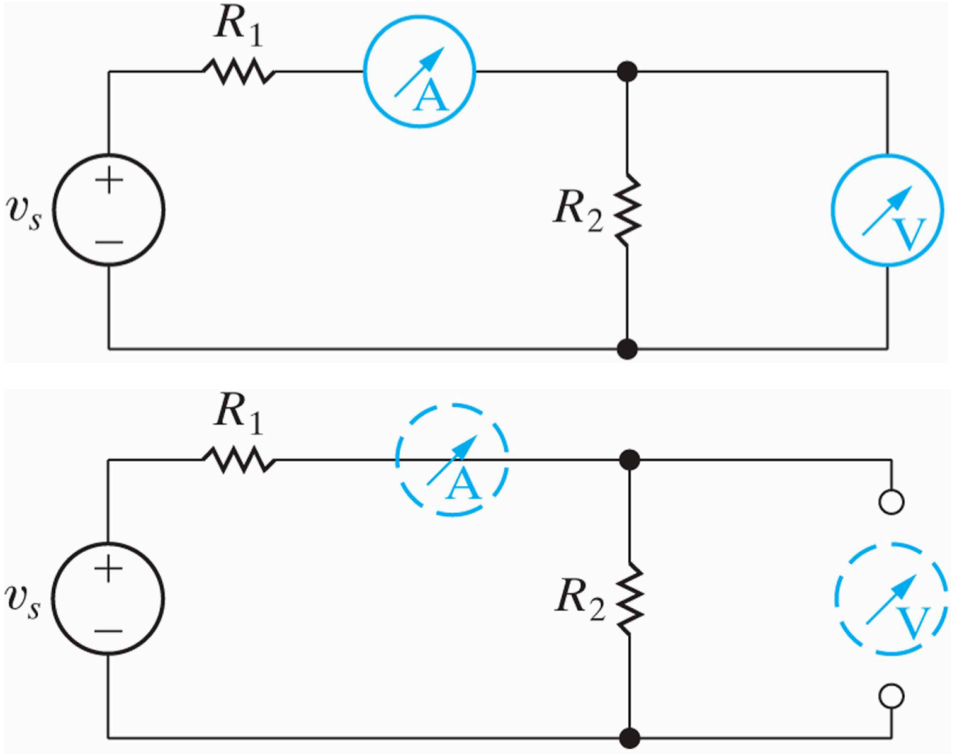
$i_1 = i\,R_p / 30 = 0.5\times10 / 30$ A $= 166.67$ mA

For (c), we have

$i_2 = i\,R_p / 60 = 0.5\times10 / 60$ A $= 1/12$ A

$p_2 = 50 i_2^2 = 50 / 144$ W $= 347.22$ mW

Measuring Voltage and Current



An **ammeter** is an instrument designed to measure current;
A **voltmeter** is an instrument designed to measure voltage

An **ideal ammeter** has an equivalent resistance of 0 Ω and functions as a short circuit in series with the element whose current is being measured.
An **ideal voltmeter** has an infinite equivalent resistance and thus functions as an open circuit in parallel with the element whose voltage is being measured.

Digital and Analog Meters



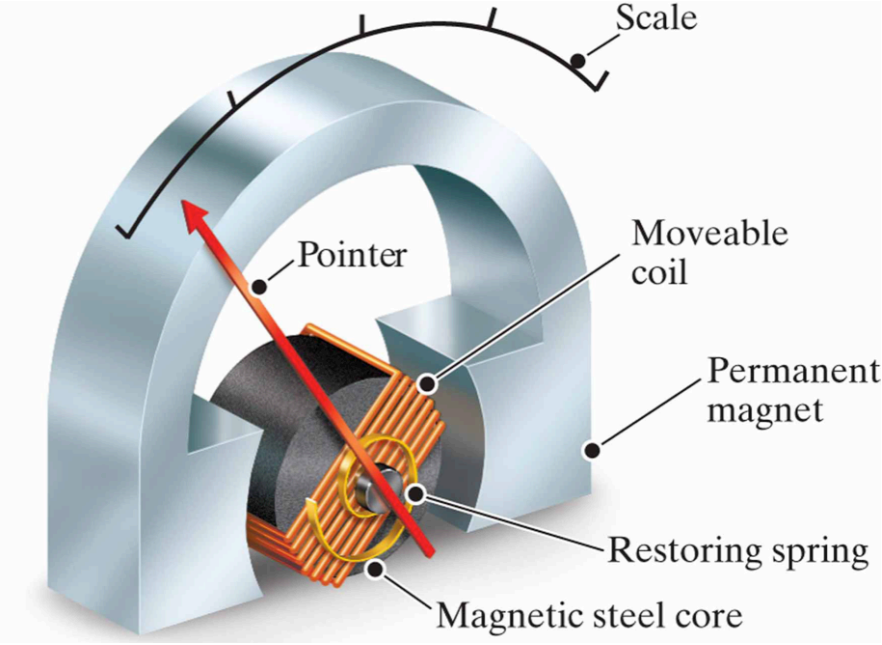






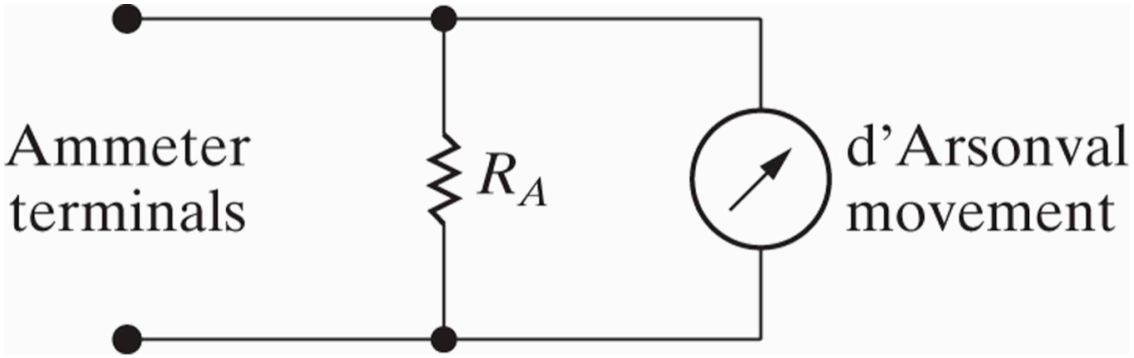
d'Arsonval Meter Movement

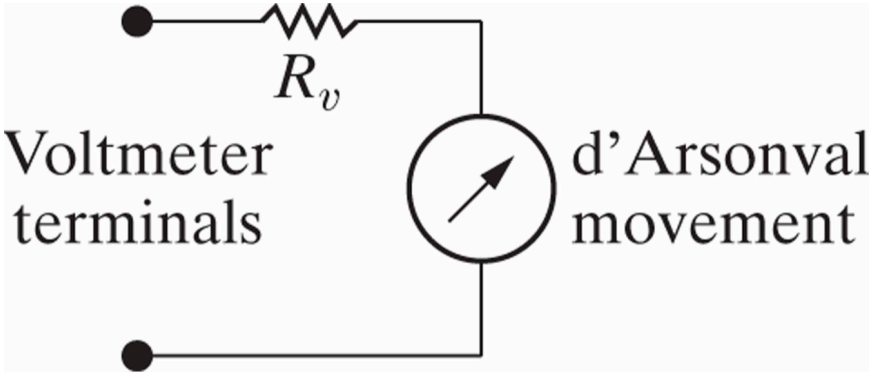
d'Arsonval meter movement consists of a movable coil placed in the field of a permanent magnet. When current flows in the coil, it creates a torque on the coil, causing it to rotate and move a pointer across a calibrated scale. *By design, the deflection of the pointer is directly proportional to the current in the movable coil.* The coil is characterized by both a voltage rating and a current rating.



For example, one commercially available meter movement is rated at 50 mV and 1 mA. This means that when the coil is carrying 1 mA, the voltage drop across the coil is 50 mV and the pointer is deflected to its full-scale position.

DC Ammeter/Voltage Circuit



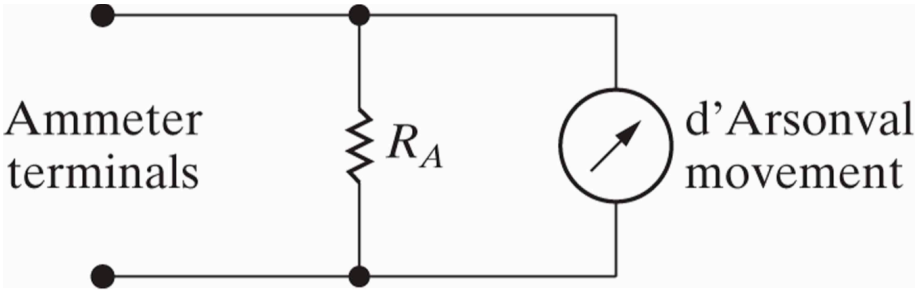


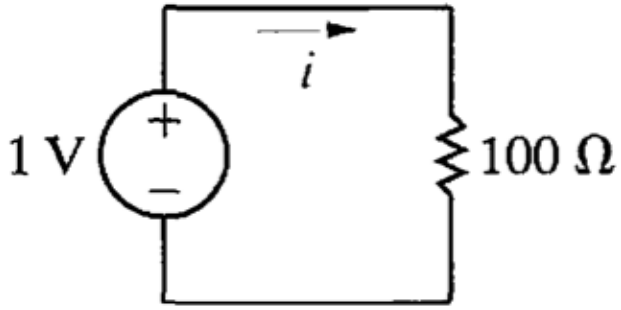
Example #4

a) A 50 mV, 1 mA d'Arsonval movement (left) is to be used in an ammeter with a full-scale reading of 10 mA. Determine R_A.

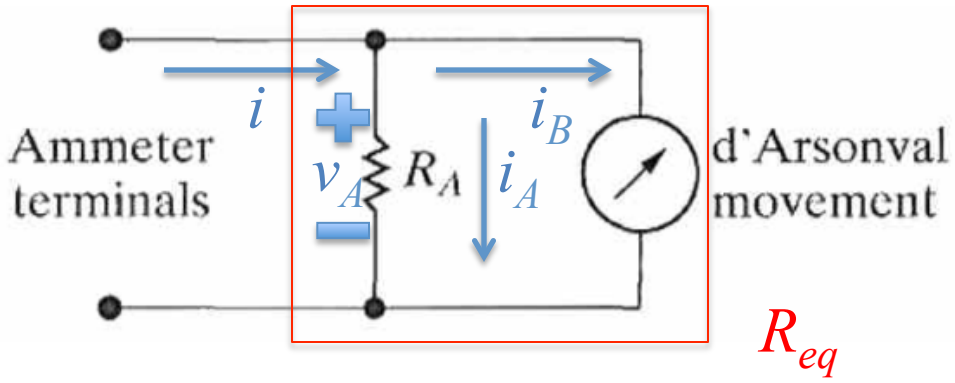
b) Find the current in the circuit shown (right).

c) If the ammeter (left) is used to measure the current in the circuit (right), what will it read?





Solution for Example #4



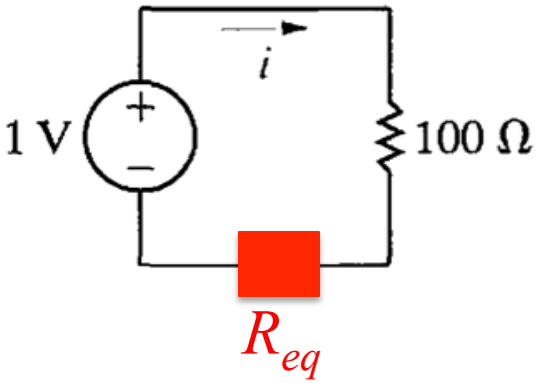
For (a) (left), we have

$i_A = I - i_B = 10\text{ mA} - 1\text{mA} = 9\text{ mA}$

$R_A = v_A / i_A = 50 / 9 = 5.56\ \Omega$

For (b) (right), we have

$i = 1 / 100\text{ A} = 10\text{ mA}$

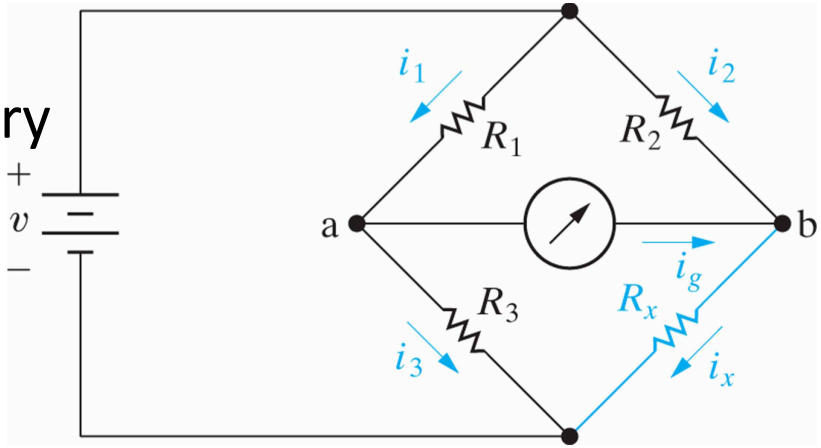


For (c), we have

$R_{eq} = v_A / i = 50 / 10 = 5\ \Omega$

$i = 1 / (100 + R_{eq}) = 1 / 105\text{ A} = 9.524\text{ mA}$

Measuring Resistance: The Wheatstone Bridge



$$i_g = 0$$

$$i_1 = i_3 \quad i_2 = i_x$$

$$i_1 R_1 = i_2 R_2 \quad i_3 R_3 = i_x R_x$$

$$R_1 / R_2 = i_2 / i_1$$

$$R_3 / R_x = i_x / i_3 = i_2 / i_1$$

In a commercial Wheatstone bridge, R_1 and R_2 consist of decimal values of resistances that can be switched into the bridge circuit.

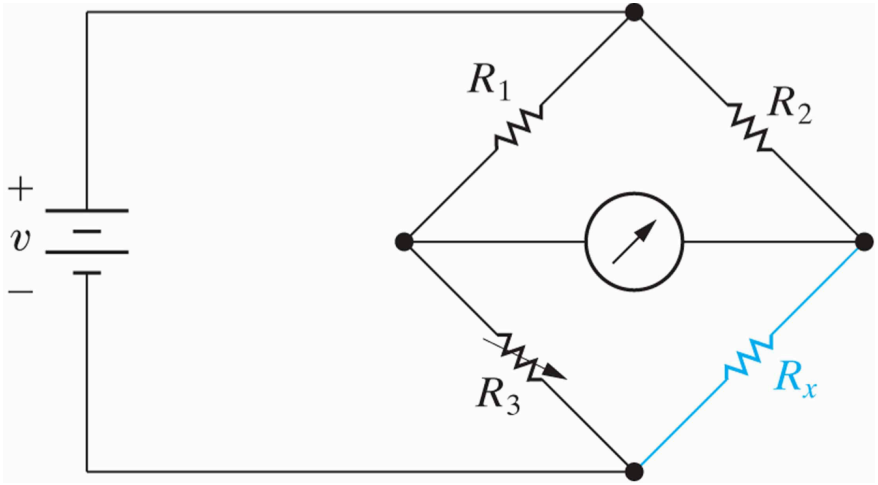
$$\boldsymbol{R_x = R_2 R_3 / R_1}$$

Example #5

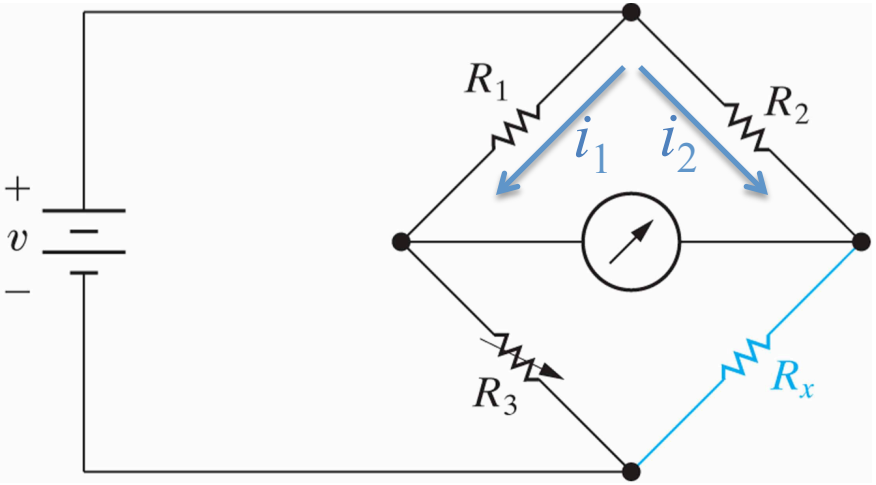
The bridge circuit shown is balanced when R_1=100 Ω, R_2 =1000 Ω, and R_3 =150 Ω. The bridge is energized from a 5 V dc source.

a) What is the value of R_x?

b) Suppose each bridge resistor is capable of dissipating 250 mW. Can the bridge be left in the balanced state without exceeding the power-dissipating capacity of the resistors, thereby damaging the bridge?



Solution for Example #5



$$p_1 = i_1^2 R_1 = 0.02^2 \times 100 = 0.04 \text{ W}$$
$$p_2 = i_2^2 R_2 = 0.002^2 \times 1000 = 0.004 \text{ W}$$
$$p_3 = i_1^2 R_3 = 0.02^2 \times 150 = 0.06 \text{ W}$$
$$p_x = i_2^2 R_x = 0.002^2 \times 1500 = 0.006 \text{ W}$$

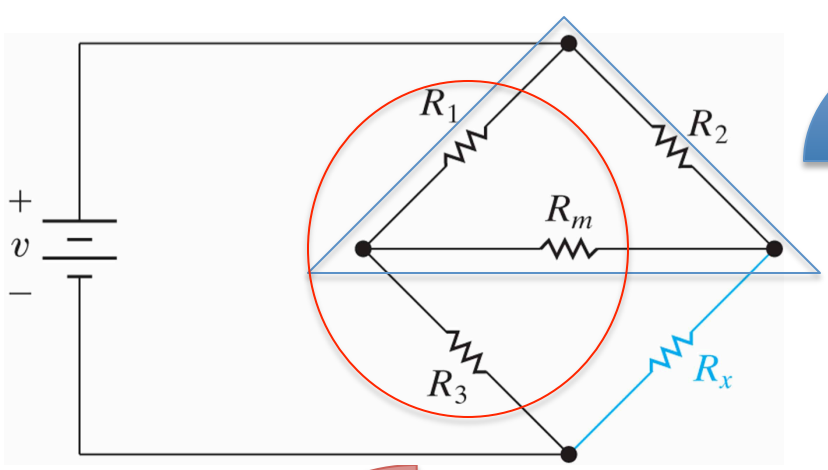
For (a), we have

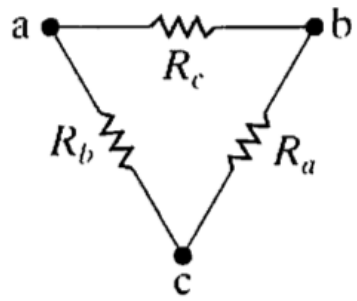
$$R_x = R_2 R_3 / R_1 = 1000 \times 150 / 100 = 1500$$

For (b), we have

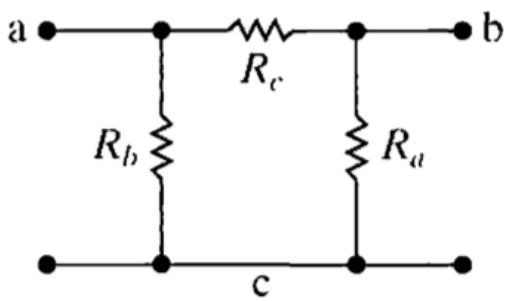
$$i_1 = v / (R_1 + R_3) = 5 / 250 = 0.02 \text{ A}$$
$$i_2 = v / (R_2 + R_x) = 5 / 2500 = 0.002 \text{ A}$$

Delta-to-Wye (Pi-to-Tee) Equivalent Circuits

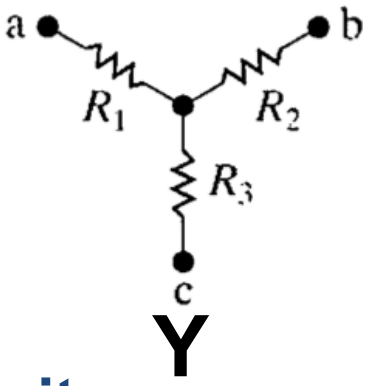




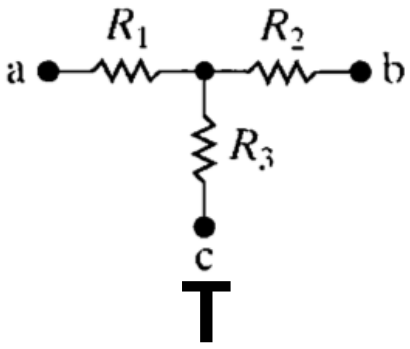
Δ



π

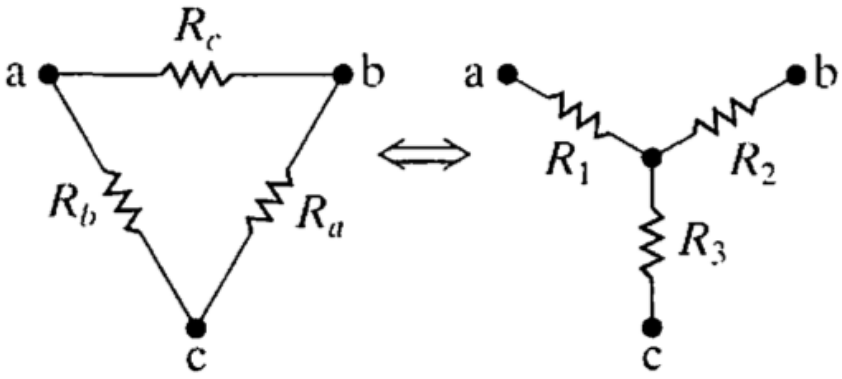


Y



T

The Δ-to-Y Transformation



$$R_{ab} = \frac{R_c(R_a + R_b)}{R_a + R_b + R_c} = R_1 + R_2$$

$$R_{bc} = \frac{R_a(R_b + R_c)}{R_a + R_b + R_c} = R_2 + R_3$$

$$R_{ca} = \frac{R_b(R_c + R_a)}{R_a + R_b + R_c} = R_1 + R_3$$

$$R_1 = (R_{ab}+R_{ca}-R_{bc}) / 2 = R_bR_c / (R_a+R_b+R_c)$$
$$R_2 = (R_{ab}+R_{bc}-R_{ca}) / 2 = R_cR_a / (R_a+R_b+R_c)$$
$$R_3 = (R_{bc}+R_{ca}-R_{ab}) / 2 = R_aR_b / (R_a+R_b+R_c)$$

How about Y-to-Δ?

The Y-to-Δ Transformation

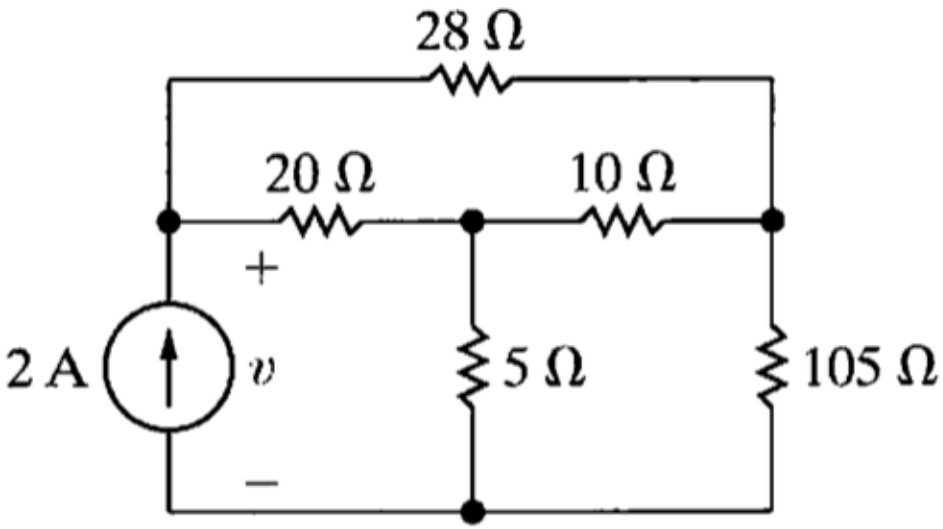
$$R_a = \frac{R_1R_2 + R_2R_3 + R_3R_1}{R_1},$$

$$R_b = \frac{R_1R_2 + R_2R_3 + R_3R_1}{R_2},$$

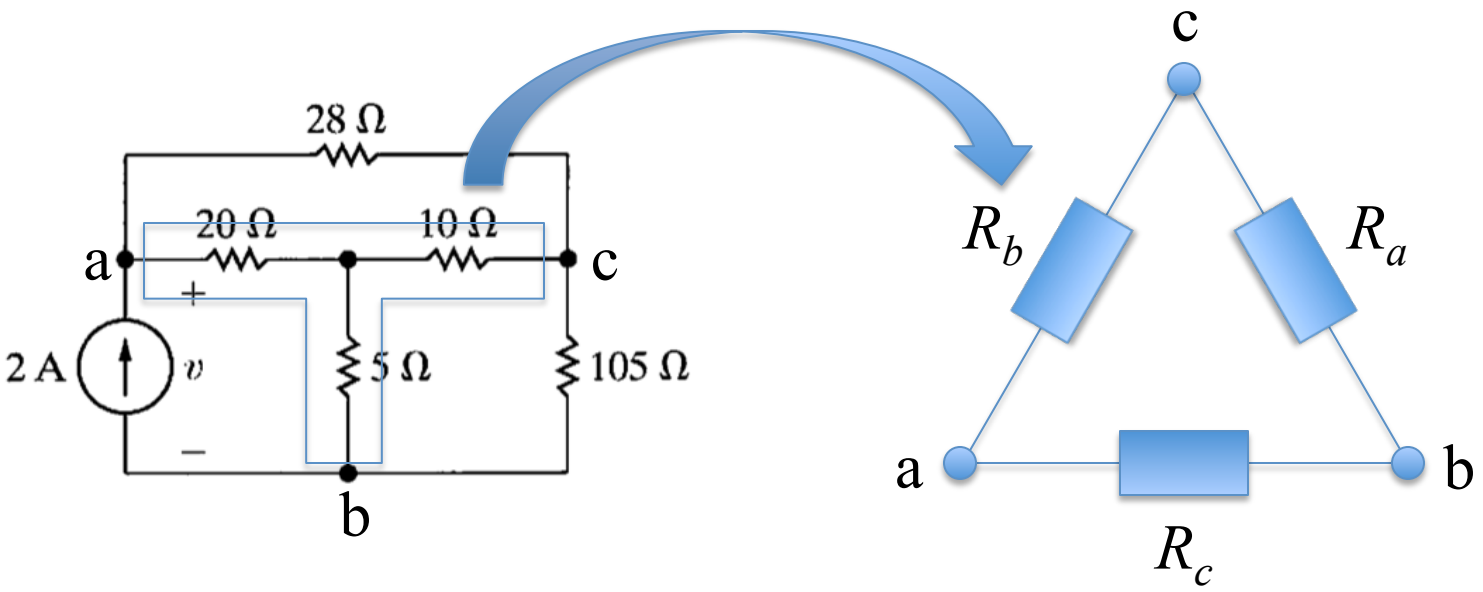
$$R_c = \frac{R_1R_2 + R_2R_3 + R_3R_1}{R_3}.$$

Example #6

Use a Y-to-Δ transformation to find the voltage v in the circuit shown.



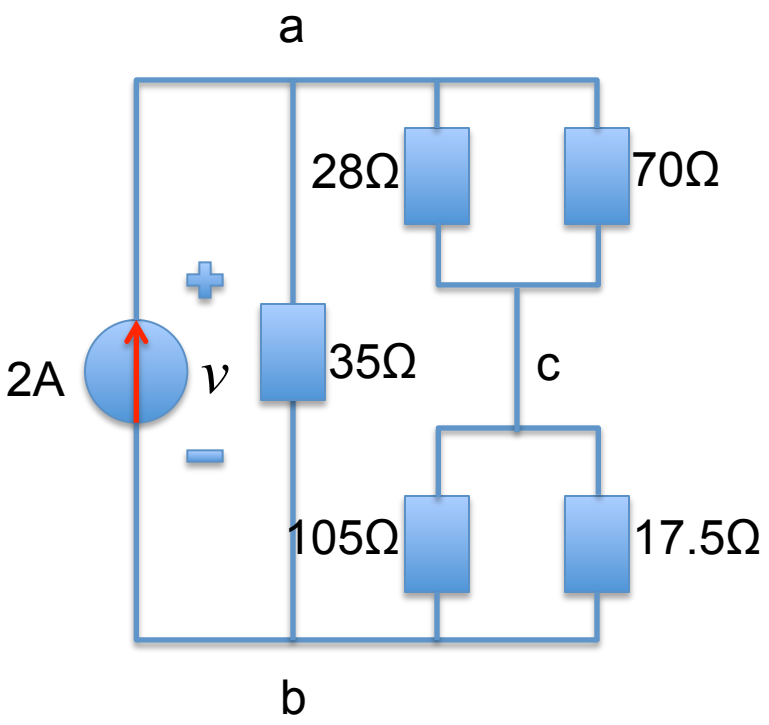
Solution for Example #6



$$R_a = (20\times10+20\times5+10\times5) / 20 = 17.5\ \Omega$$

$$R_b = (20\times10+20\times5+10\times5) / 5 = 70\ \Omega$$

$$R_c = (20\times10+20\times5+10\times5) / 10 = 35\ \Omega$$



$$R_{eq} = 35||[(28||70)+105||17.5]$$
$$= 35||(20+15)$$
$$= 17.5\ \Omega$$

$$v = 2R_{eq} = 35\ \text{V}$$

Summary

- Series/Parallel resistors
- Voltage/Current division
- Digital/Analog voltmeter/ammeter
- Wheatstone bridge
- Δ-to-Y/Y-to-Δ transformation